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NIXON & VANDERHYE, PC
901 NORTH GLEBE ROAD, 11TH FLOOR
ARLINGTON, VA 22203

EXAMINER
DIXON, ANNETTE FREDRICKA

ART UNIT	PAPER NUMBER
3785	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOMAIL@nixonvan.com
pair_nixon@firsttofile.com

Office Action Summary	Application No. 17/255,615	Applicant(s) BARLOW et al.	
	Examiner ANNETTE F DIXON	Art Unit 3785	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 4/27/2022.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-20 and 75-77 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-20 and 75-77 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 4/27/2022.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. This Office Action is in response to the amendment, filed on April 27, 2022. Primary Examiner acknowledges Claims 1-20 and 75-77 are pending in this application, with Claim 1 having been currently amended, and Claims 21-74 having been cancelled.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 102

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

4. Claims 1, 3, 5-12, 15-20, and 75-77 are rejected under 35 U.S.C. 102(a)(1) as being clearly anticipated by Ovzinsky (9,731,090).

The applied reference has a common assignee (ResMed) with the instant application. Based upon the earlier effectively filed date of the reference, it constitutes prior art under 35 U.S.C. 102(a)(1). Primary Examiner notes the effectively filed date of this instant application is March 28, 2019; however, the effective filing date of this prior art has a patent date of August 15, 2017 which is before the one (1) year grace period.

As to Claim 1, Ovzinsky discloses a patient interface (Figure 6), comprising: a plenum chamber (defined by the region of 510 remote from the patient's nasal region, "FIG. 6 shows an example of a pair of air delivery conduits communicated with the patient interface. As illustrated, a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Column 7, Lines 30-40) pressurizable to a therapeutic pressure of at least 6 cm of water above ambient pressure ("the PAP device generates a supply of pressurized air (e.g., 2-30 cm H.sub.2O) that is delivered to the patient interface via the air delivery conduit." Column 6, Lines 1-10), said plenum chamber (defined by the region of 510 remote from the patient's nasal region) including a plenum chamber inlet port (520(1), "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement)" Column 7, Lines 30-40) sized and structured to receive a flow of air at the therapeutic pressure for breathing by the patient, a seal forming structure (defined by the region of 510 adjacent and inserted into the patient's nasal region via "e.g., nasal prong/nozzle arrangement" Column 7, Lines 30-40) constructed and arranged to form a seal with a region of the patient's face surrounding an entrance to the patient's airways, the seal forming structure (defined by the region of 510 adjacent and inserted into the patient's nasal region) having a hole therein such that the flow of air at said therapeutic pressure is delivered to at least an entrance to the patient's nares, the seal forming structure (defined by the region of 510

adjacent and inserted into the patient's nasal region) constructed and arranged to maintain said therapeutic pressure in the plenum chamber throughout the patient's respiratory cycle in use; a positioning and stabilizing structure (defined by the cooperative interactions of 515, 520, and 545) to provide a force to hold the seal forming structure (defined by the region of 510 adjacent and inserted into the patient's nasal region) in a therapeutically effective position on the head of the patient, the positioning and stabilizing structure (defined by the cooperative interactions of 515, 520, and 545) comprising: at least one gas delivery tube (520, "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit. In this example, each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Column 7, Lines 30-40) to deliver the flow of air to the entrance of the patient's airways via the seal forming structure (defined by the region of 510 adjacent and inserted into the patient's nasal region), the at least one gas delivery tube (520) being constructed and arranged to contact, in use, at least a region of the patient's head superior to an otobasion superior of the patient's head ("each conduit 520 is adapted to extend from adjacent to or under the patient's nose, over the patient's cheeks, between the patient's eye and ear, and terminate at the crown of the patient's head." Column 7, Lines 30-40); wherein the at least one gas delivery tube (520) includes an inner textile layer (one of 655 defined as the portion of the gas delivery conduit x20 proximate to the patient's face, as shown in Figures 10-15 in which each of 621 and 622 form

the entire gas delivery conduit 620 or x20, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit." Column 12, Lines 30-45) and an outer textile layer (other of 655 defined as the portion of the gas delivery conduit x20 remote to the patient's face), the inner textile layer (one of 655) being joined to the outer textile layer (other of 655) along a first edge of the gas delivery tube (x20) thereby forming a first seam (defined as one of the regions upon which 621 and 622 are attached) such that the inner textile layer (one of 655) is spaced from the outer textile layer (other of 655) along a central portion of the gas delivery tube (x20) to form a tubular shaped gas passageway to convey the flow of air, wherein the first seam (defined as one of the regions upon which 621 and 622 are attached) closes off a lateral side of the gas passageway to prevent the escape of the flow of air to the atmosphere, and the first seam (defined as one of the regions upon which 621 and 622 are attached) provides rigidity to the gas delivery tube (x20).

Regarding the shape of the gas delivery tube portions, Primary Examiner notes Ovzinsky explicitly discloses "In the illustrated example, the first and second conduit portions are symmetrical, however it should be appreciated that the conduit portions may be asymmetrical." (Column 12, Lines 40-45). Consequently, it is understood if the conduit portions (x21 and x22) of the gas delivery tube (x20) are "asymmetrical" the cross-sectional shape of these conduit portions (x21 and x22) would be different whereby one shape would be more planar than the other and this configuration would effectuate the operational characteristics of the gas delivery tube. Consequently, based in this disclosure of the "asymmetrical" cross sectional shape, Ovzinsky discloses the inner textile layer (one of 655 which extends beyond the

midpoint of symmetry of the tubular gas delivery tube) has a substantially planar cross sectional shape, wherein the outer textile layer (other of 655 which is complementary the remaining shape of the tubular gas delivery tube) has a predetermined non-planar cross sectional shape.

Regarding the limitations to "wherein the inner textile layer has a substantially planar cross-sectional shape from a first lateral edge of the inner textile layer to a second lateral edge of the inner textile layer, wherein the outer textile layer has a pre-determined non-planar cross-sectional shape", Primary Examiner notes Ovzinsky discloses the gas delivery tube may be constructed of "other suitable cross-sectional shapes are possible, e.g., generally D-shaped cross-section." (Column 11, Lines 60-65); and "other suitable cross-sectional shapes are possible, e.g., cylindrical, generally D-shaped cross-section. In an example, as shown in FIG. 21, d1 along the major axis is about 15-25 mm, e.g., 18.9 mm or 19 mm, and d2 along the minor axis is about 5-15 mm, e.g., 9.4 mm or 9 mm." (Column 13, Lines 25-35). In this disclosed configuration of a "D-shaped cross-section" it is understood a configuration of the gas delivery tube having the inner textile layer and the outer textile layer wherein the inner textile layer has a planar cross section of the upright of the "D"; while the outer textile layer possesses the predetermined non-planar cross section of the curvature of the "D" meets the claimed configuration.

Regarding the limitations to "wherein the outer textile layer is resilient such that when subjected to external force the cross- sectional shape of the outer textile layer is altered, and when the external force is released the outer textile layer returns to the pre-determined non-planar cross-sectional shape", Primary Examiner notes Ovzinsky discloses "In use, such a conduit keeps its form in the absence of external force. However, the conduit is also collapsible

and, when sufficient external force is applied, the conduit collapses. The form holding and collapsible features are particularly useful in the configuration shown in FIG. 6 which includes two conduits 520, one on each side of the patient's head. While both conduits maintain their form in the absence of external force, when the patient is lying in bed and turns to one side, the respective conduit conveniently collapses, thus improving patient comfort. The remaining conduit maintains the air supply to the patient interface 510. The conduit may also be air impermeable." (Column 12, Line 55 thru Column 13, Line 5).

Regarding the seam language, Ovzinsky explicitly discloses "the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. During the thermoforming process, the laminate adheres to the fabric giving it air impermeable properties. The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 50-60).

Thus, it is clear that Ovzinsky discloses all of the elements of the claimed invention whereby there is a plenum chamber, a seal forming structure, a positioning and stabilizing structure, and a series of tube configurations which permit for collapsible resiliency in the face of external forces.

As to Claim 3, Ovzinsky discloses the at least one gas delivery tube (x20) further includes an inner textile membrane (x50, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a

polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55) provided to the inner textile layer (x55, "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12)" Column 12, Lines 50-55), wherein during use the inner textile membrane (x50) is configured to resist the transfer of air through the inner textile layer (x55).

As to Claim 5, Ovzinsky discloses an inner textile layer (x55 one of 621/622) is joined to an outer textile layer (x55 other of 621/622) along a second edge (Figure 13 and 14) of the gas delivery tube (x20) thereby forming a second seam ("seam welded" Column 12, Lines 50-60) of the gas delivery tube (x20).

As to Claim 6, Ovzinsky discloses the inner textile layer (x55) is floppy ("In use, such a conduit keeps its form in the absence of external force. However, the conduit is also collapsible and, when sufficient external force is applied, the conduit collapses. The form holding and collapsible features are particularly useful in the configuration shown in FIG. 6 which includes two conduits 520, one on each side of the patient's head. While both conduits maintain their form in the absence of external force, when the patient is lying in bed and turns to one side, the respective conduit conveniently collapses, thus improving patient comfort. The remaining conduit maintains the air supply to the patient interface 510. The conduit may also be air impermeable." Column 12, Line 55 thru Column 13, Line 5).

As to Claim 7, Ovzinsky discloses the inner textile layer (x55) is secured to (best seen Figure 14) the outer textile layer (x55), wherein the outer textile layer (x55) imparts tension on the inner textile layer (x55).

As to Claim 8, Ovzinsky discloses the inner textile layer (x55) forms an exterior textile sheet (best seen Figures 10-14), wherein the inner textile membrane (x50) is provided to the inner surface of the inner textile layer (x55), wherein the inner textile membrane (x50) is configured to resist the transfer of air through the exterior textile sheet. ("Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55).

As to Claim 9, Ovzinsky discloses the at least one gas delivery tube (x20) is configured to extend continuously from a right side of the patient, along the parietal bone, to the left side of the patient (through the manifold 515, best seen Figure 6, "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Column 7, Lines 30-40).

As to Claim 10, Ovzinsky discloses at least one gas delivery tube (x20) includes an air supply conduit (515, "a first cuff or end 520(1) of each conduit 520 may be adapted to engage a

respective end or inlet of the patient interface 510 (e.g., nasal prong/nozzle arrangement) and the second cuff or end 520(2) may be adapted to engage a respective end of a manifold 515 communicated with the outlet of the PAP device via another air delivery conduit." Column 7, Lines 30-40).

As to Claim 11, Ovzinsky discloses the outer textile layer (x55) forms an exterior textile sheet (best seen Figures 10-14), wherein the outer textile membrane (x50) is provided to the inner surface of the outer textile layer (x55), wherein the conformable sheet is sandwiched between the exterior textile sheet and the outer textile membrane (x50).

As to Claim 12, Ovzinsky discloses a width of the first seam ("seam welded" Column 12, Lines 50-60) is varied along a length of the at least one gas delivery tube (x20).

As to Claim 15, Ovzinsky discloses the inner textile layer (x55) and the outer textile layer (x55) are joined together without stitching - "The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 50-60).

As to Claim 16, the modified Ovzinsky, specifically Ovzinsky discloses the at least one gas delivery tube (x20) having a conformable inner textile membrane (x50 on one of 621/622, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical

complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55) provided to the inner textile layer (x55), the inner textile membrane (x50) being air impermeable, and a conformable outer textile membrane (x50 on other of 621/622, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55) provided to the outer textile layer (x55), the outer textile layer (x55) being air impermeable. Further, Ovzinsky discloses the inner textile layer (x55) and the outer textile layer (x55) are joined together without stitching - "The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 50-60).

As to Claim 17, Ovzinsky discloses the inner textile layer (x55) and the outer textile layer (x55) are joined together - The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 50-60). Ovzinsky is absent to the statement of "rigidizers" thus, the claimed limitation of "rigidizers not utilized" is inherent.

As to Claim 18, Ovzinsky discloses "In use, such a conduit keeps its form in the absence of external force. However, the conduit is also collapsible and, when sufficient external force is applied, the conduit collapses. The form holding and collapsible features are particularly useful in the configuration shown in FIG. 6 which includes two conduits 520, one on each side of the patient's head. While both conduits maintain their form in the absence of external force, when the patient is lying in bed and turns to one side, the respective conduit conveniently collapses, thus improving patient comfort. The remaining conduit maintains the air supply to the patient interface 510. The conduit may also be air impermeable." (Column 12, Line 55 thru Column 13, Line 5). Consequently the configuration upon which there is the least external force will facilitate the movement of the inner textile layer (x55) away from the outer textile layer (x55).

As to Claim 19, Ovzinsky discloses "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." (Column 12, Lines 50-55). Consequently, the gas delivery tube (x20) is constructed with conformable materials (x50 - "film laminate") upon which the outer textile layer (x55) is thermoformed to non-planar cross sectional shape of the outer textile layer (x55).

As to Claim 20, please see the rejection of Claim 1 which addresses the concept of the external force. Ovzinsky discloses "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." (Column 12, Lines 50-55). By this configuration the claimed orientations of "orthogonal" are considered.

As to Claims 75-77, Ovzinsky discloses "The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622. … The two conduit portions 621 and 622 can then be seam welded as shown in FIG. 13 (e.g., RF weld to couple the conduit portions) and then ultrasonically die cut as shown in FIG. 14 (e.g., to remove seam edges) to create an air tight textile conduit." (Column 12, Lines 50-65). As addressed Ovzinsky discloses the "the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric." (Column 12, Lines 45-50). Still further, related to Claim 76, Ovzinsky discloses "the air delivery conduit may be dip molded, e.g., use silicone, polyurethane or synthetic polyisoprene" are functionally equivalent materials. (Column 8, Lines 30-40).

Claim Rejections - 35 USC § 103

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. Claims 2, 4, 13, and 14 are rejected under 35 U.S.C. 103 as being obvious over Ovzinsky (9,731,090) in view of Meyer et al. (2005/0284481).

The applied reference has a common assignee (ResMed) with the instant application. Based upon the earlier effectively filed date of the reference, it constitutes prior art under 35 U.S.C. 102(a)(1). Primary Examiner notes the effectively filed date of this instant application is

March 28, 2019; however, the effective filing date of this prior art has a patent date of August 15, 2017 which is before the one (1) year grace period.

As to Claim 2, Ovzinsky discloses the configuration of an inner textile layer (one of 655) and an outer textile layer (other of 655) for the construction of a gas delivery tube (x20) wherein the gas delivery tube (x20) is constructed of two parts (x21 and x22).

Yet, Ovzinsky **does not expressly disclose** "wherein the inner textile layer has different material properties than the outer textile layer."

Meyer teaches a similar construction of a gas delivery tube for the application of supplying pressurized gas to the patient interface (Abstract and Figures 2 and 3), wherein a source of gas is supplied (via 10, "a CPAP respirator" Para 0021-0022) and further conveyed through a gas delivery tube (5/6, "The breathing gases fed in pass through a channel 5 integrated in the strap in FIG. 1 and enter the interior of the mask body from there. The breathing gases fed in may also pass through a channel 6 which is likewise integrated within the strap." Para 0020) to the patient interface (1, "FIG. 1 shows a CPAP mask designed as a nose mask, whose mask body comprises a rigid frame 1." Para 0020) having a chamber region (3, "the frame 1 is provided with a moldable support frame 3 in the form of a grid-like skeleton, which approximately predetermines the shape of the mask body." Para 0020) whereby a seal forming structure (2, "this frame 1 carries a flexible, very soft seal 2, which is supported by the internal pressure, which is provided by the CPAP respirator." Para 0020) is supplied to encompass the nose of the patient (best seen Figures 2 and 3). Further, Meyer teaches the configuration of the gas delivery tubes (5/6) as "an at least two-layer textile fabric, in which the layers are connected with one another such that they form longitudinally extending channels."

(Para 0009). Finally, Meyer's teaches the gas delivery tube construction for the same purpose of "external force" response having an expanded configuration and a non-expanded configuration as best seen in Figure 1 - "FIG. 1 illustrates the two pathways with channel 6 shown in the non-unfolded (not expanded) state, which may be due, for example, to the head lying on a bed." (Para 0020). Consequently, the configuration of the gas delivery tube is a function of the pneumatic resistance whereby variations in thickness and number of fabric layers may effect the characteristics of the gas delivery tube. Thus, an external force such due to "the head lying on a bed" would effectively prevent the ability of the outer textile layer to expand (see Figure 1, tube 6 at the location remote from the patient's face); however, once that external force is released, the gas delivery tube would be able to expand (see Figure 1, tube 5 at the location remote from the user's face). Thus the resultant effect of this variable configuration based on external forces is the "increased wearing comfort and increased acceptance by the user and his social environment." (Para 0007) such that "the patient does not have to pay attention to the position in which he holds his head." (Para 0011)

With respect to the material composition of the gas delivery tubes, Meyer teaches "The pneumatic resistance can be set for different conditions of use by using different mesh sizes, fabric thicknesses and a variable number of fabric layers." (Para 0020), wherein "individual areas with different numbers of layers can be formed, which have different pneumatic resistances, as a result of which the discharge geometry can be affected by the mask body. The flow resistance can thus be adapted to the standby pressure, which is preferably set, by varying the number of layers. It is also possible to form zones with different flow resistances by combining different fabric structures." (Para 0012).

Therefore, it would have been obvious to one having ordinary skill in the art to construct the gas delivery tubes of Ovzinsky to have different material properties between the inner textile layer and the outer textile layer, as taught by Meyer to control the "discharge geometry" and create "zones with different flow resistances".

As to Claim 4, the modified Ovzinsky, specifically Ovzinsky discloses the at least one gas delivery tube (x20) having a conformable inner textile membrane (x50 on one of 621/622, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55) provided to the inner textile layer (x55), the inner textile membrane (x50) being air impermeable, and a conformable outer textile membrane (x50 on other of 621/622, "Each conduit portion 621, 622 includes an inner layer of a film laminate 650 that forms an interior surface of the conduit and an outer layer of a textile or fabric 655 that forms an exterior surface of the conduit. In an example, the film laminate is a polyurethane or medical grade film, and the textile is a thermoformable fabric. The film laminate 650 is applied to the fabric 655 (FIG. 11), and then the fabric and laminate are both thermoformed to create symmetrical complementary shapes (FIG. 12), i.e., conduit portions 621, 622." Column 12, Lines 40-55) provided to the outer textile layer (x55), the outer textile layer (x55) being air impermeable.

Yet, Ovzinsky **does not expressly disclose** the "outer textile layer is thicker than the inner textile layer".

Regardless this configuration is obvious to try from a finite number of predictable results as a function of the teachings of Myer, which discloses the variations in pneumatic resistance can result from "different conditions of use by using different mesh sizes, fabric thicknesses and a variable number of fabric layers." (Para 0020, also see 0012). Consequently, the decision to have the outer textile layer thicker than the inner textile layer has a known result effective variable of providing a different region having a different flow resistance.

Moreover, Applicant has not asserted the specific construction provides a particular advantage, solves a stated problem or serves a purpose different from that of providing configuration suitable for the passage of gases; thus, the specific construction lacks criticality in its design Consequently, one of ordinary skill in the art would have expected Applicant's invention to perform equally well with the modified Ovzinsky, as the construction would yield the predictable results of providing a gas delivery tube having variable zones of flow resistance. Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube to have an outer layer thicker than the inner layer, a known result effective suitable for providing variations of flow resistance along the gas delivery tube.

As to Claims 13 and 14, the modified Ovzinsky discloses the configuration of an "asymmetrical" the cross-sectional shape of these conduit portions (x21 and x22) would be different whereby one shape would be more planar than the other and this configuration would effectuate the operational characteristics of the gas delivery tube. Consequently, based

in this disclosure of the "asymmetrical" cross sectional shape, Ovzinsky discloses the inner textile layer (one of 655 which extends beyond the midpoint of symmetry of the tubular gas delivery tube) has a substantially planar cross sectional shape, wherein the outer textile layer (other of 655 which is complementary the remaining shape of the tubular gas delivery tube) has a predetermined non-planar cross sectional shape.

Yet, Ovzinsky **does not expressly disclose** the curvatures of each of the outer textile layer and the inner textile layer. As shown in Meyers the curvature of the greater inner textile layer would have a flatter orientation along the head of the wearer, than the outer textile membrane which would have a positive curvature to permit the expansion of the gas delivery tubes as seen in Figures 5 and 6).

Therefore, it would have been obvious to one having ordinary skill in the art to modify the construction of the gas delivery tube to have an outer textile layer with a positive curvature and the inner textile layer with no curvature, a known result effective suitable for providing variations of flow resistance along the gas delivery tube.

Response to Arguments

7. Applicant's arguments with respect to claims have been considered but are not persuasive to overcome the prior art made of record despite the further amendment to the claims.

Although Primary Examiner appreciates Applicant's clarifying amendment incorporating "wherein the inner textile layer has a substantially planar cross-sectional shape from a first lateral edge of the inner textile layer to a second lateral edge of the inner textile layer, wherein

the outer textile layer has a pre-determined non-planar cross-sectional shape", it should be noted that the prior art made of record already disclosed the newly added features.

Explicitly, Ovzinsky discloses the gas delivery tube may be constructed of "other suitable cross-sectional shapes are possible, e.g., generally D-shaped cross-section." (Column 11, Lines 60-65); and "other suitable cross-sectional shapes are possible, e.g., cylindrical, generally D-shaped cross-section. In an example, as shown in FIG. 21, d1 along the major axis is about 15-25 mm, e.g., 18.9 mm or 19 mm, and d2 along the minor axis is about 5-15 mm, e.g., 9.4 mm or 9 mm." (Column 13, Lines 25-35). In this disclosed configuration of a "D-shaped cross-section" it is understood a configuration of the gas delivery tube having the inner textile layer and the outer textile layer wherein the inner textile layer has a planar cross section of the upright of the "D"; while the outer textile layer possesses the predetermined non-planar cross section of the curvature of the "D" meets the claimed configuration.

In light of the aforementioned reasoning, Primary Examiner has maintained this rejection and made it FINAL.

Conclusion

8. Applicant's amendment necessitated the new ground(s) of rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of

the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to ANNETTE F DIXON whose telephone number is (571)272-3392. The examiner can normally be reached M-F 9-5 EST with flexible hours.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Kendra D Carter can be reached on 571-272-9034. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO

Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

ANNETTE FREDRICKA DIXON
Primary Examiner
Art Unit 3782

/Annette Dixon/
Primary Examiner, Art Unit 3785